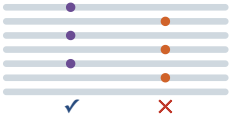
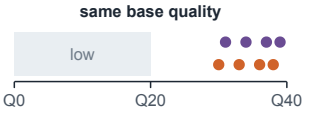
A Where the background comes from

Quality-based test (e.g. LoFreq, Geneious)

✓ correct ✕ wrong



mutation called — error miscalled

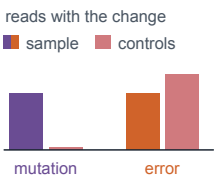


errors from PCR, homopolymers, paralog reads look like mutations

Control-calibrated test (SHMtrace)



mutation called — error rejected



AID-null controls from the same run; estimated FDR ≤ 0.05

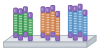
B Inputs and read preparation



Nanopore full-length amplicons



Sanger .ab1 traces + sample sheet

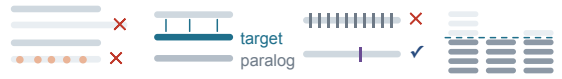


Illumina paired reads, merged



UMI-tagged one consensus per molecule

Read preparation



over-length or > 1% errors/base — aligned to both, best match kept — ≥ 10 paralog bases: removed — random draw, same per library

Sanger: trimming at Q ≥ 20 replaces 1; every clone is kept, so no 4

C Calling: a test for reads, a count for Sanger clones

Reads (nanopore, Illumina, UMI): tested against controls

1 Control error model



rate per type, strand, quality + held-out margin

2 Test and FDR



decoys (open) set the threshold: FDR ≤ 0.05

Sanger clones: counted

1 Count the mismatches



every substitution at quality ≥ 20 is counted

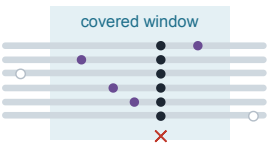
2 Background reported



shown beside the calls, never subtracted

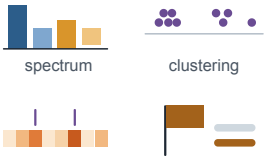
Both routes

3 Germline and window



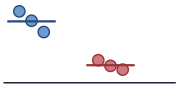
germline (≥ 50% of reads) out; calls inside the window only; indels optional

4 Automatic checks



D Outputs

Animal-level tests and power



NB, Welch, permutation

Single-read estimate



observed minus control background

Clones from the V(D)J junction



reads grouped by junction; each clone counted once

Mutation trees



full-length reads of one clone

Spectrum and motif shares



G:C / A:T, Ts / Tv, WRC/GYW, WA/TW

Reports, app and command line



HTML report, tables, project files